

Curriculum Intelligence Engine — Week 4: Improved Model (redo)

Group 9 · Roll Nos: J048, J049, J050, J052, J053 · AI University NLP Course Project 2026 · redone 9 Oct 2026

CODE

```
Repository : https://github.com/jainamsethia/Curriculum-Intelligence-Engine
Tag        : week-4-redo
Commit     : 4dbcadbe4823a344677897483b84dae37ce3bf2a
Run        : python -m curriculum_engine.run week4      (after: run data, week2, week3; NMIMS_ZIP=<path to "B TECH.zip">)
Code       : curriculum_engine/experiments/week4.py (ablations), curriculum_engine/evaluate.py (paired intervals)
Folders    :
.gitignore
README.md
curriculum_engine/ the NLP engine (parser, data, overlap, redundancy, prerequisites, diff, Bloom, evaluation)
labels/           LLM-annotated reference labels (ids + labels per pass + consensus)
legacy/           old Week 3-4 scripts and reports (history only)
legacy_mit/       old MIT OCW scripts (not used)
reports/          report templates (numbers are placeholders)
requirements.txt
results/          metrics JSON + figures per week
submissions/      per-week submission folders
tests/            unit tests (pytest)
tools/            report builder, report checker, submission packer
```

Feedback addressed

Instructor comment	What we changed	Evidence
Week 4: "Where is the code?"	CODE block, tagged commit and code zip; every table below is produced by one command.	curriculum_engine/experiments/week4.py
Week 4: "overlapping content, missing pre-requisites, detect redundancies ... minimal progress"	Every ablation targets overlap, redundancy or missing prerequisites; side tasks (course search, summaries) from the first Week 4 are dropped.	results/week4/summary.json
Week 3: "Progress ... is minimal"	Baseline → core → improved on the same test items, with a 95% interval for each difference and an honest verdict.	results/week4/overlap.json

Progress vs previous week

Capability	Metric (test)	Week 2 baseline	Week 3 core	Week 4 improved	Verdict (95% CI of difference)
1 Overlap	Spearman	0.708	0.667	0.715 (fusion_rank_tfidf_maxsim)	improved
	ROC-AUC any overlap	0.959	0.932	0.963	improved
2 Redundancy	injected found in top 20	0.600	1.000	1.000 (bge-base)	encoder ablation
3 Prerequisites	removal recall / rename false alarm	1.000 / 1.000	0.880 / 0.120	Week 3 model kept	bge-base prerequisite re-tuning not run (time)
5 Bloom	accuracy	0.776	0.776	0.776	no clear difference

1. Overlap: encoder, aggregation and fusion

All methods are scored on the same 119 reference-labelled test pairs; differences to the Week 3 core use paired intervals that resample course families. Choices are made on the 80 validation pairs.

Method	Val Spearman	Test Spearman	Δ Spearman vs core [95% CI]	Test ROC-AUC	Δ AUC vs core [95% CI]	Verdict
TF-IDF cosine (Week 2)	0.677	0.708	+0.041 [-0.03, 0.12]	0.959	+0.027 [-0.02, 0.08]	no clear difference

MaxSim, MiniLM (Week 3 core)	0.673	0.667	+0.000 [0.00, 0.00]	0.932	+0.000 [0.00, 0.00]	(core)
Mean-pooled course vector, MiniLM	0.649	0.606	-0.061 [-0.13, 0.00]	0.891	-0.041 [-0.09, -0.00]	no clear difference
MaxSim, bge-base encoder	0.697	0.676	+0.009 [-0.07, 0.09]	0.938	+0.006 [-0.05, 0.06]	no clear difference
Best single unit pair, MiniLM	0.655	0.708	+0.040 [-0.00, 0.10]	0.959	+0.027 [-0.00, 0.06]	no clear difference
Fusion: rank average TF-IDF + MaxSim	0.708	0.715	+0.048 [0.00, 0.10]	0.963	+0.032 [0.00, 0.07]	improved
Fusion: logistic regression (probability)	0.678	0.726	+0.058 [0.02, 0.11]	0.971	+0.039 [0.01, 0.07]	improved

- Selected on validation: **fusion_rank_tfidf_maxsim** (test Spearman 0.715, ROC-AUC 0.963). It beats the Week 3 core, but it is only marginally above TF-IDF alone (0.708): for ranking whole courses, lexical overlap carries most of the signal. Verdicts: "improved" only if the interval of the difference is above zero, "worse" if below, otherwise "no clear difference".
- The logistic fusion turns the scores into a probability-like confidence for the API. On test: Brier score 0.138, expected calibration error 0.213; the calibration error shows it is not yet well calibrated (only 80 validation pairs to fit it), so the API will report it as a relative confidence.

2. Redundancy: encoder ablation

- Redundancy, injected near-duplicate units found in a programme's top 20: MiniLM 1.000, bge-base 1.000 (course-level TF-IDF in Week 2: 0.600).
- Prerequisites keep the Week 3 model (MiniLM, thresholds from validation cases).

3. Bloom level: lexicon vs lexicon + learned verbs + classifier

- Faculty K/L tags are the ground truth. The improved model looks the outcome's verb up in the lexicon and in a verb map learned from training outcomes, and falls back to a logistic regression on the sentence embedding (order chosen on validation: lexicon_then_learned_then_classifier).
- Test accuracy 0.776 vs 0.776 for the lexicon; difference 0.000 [0.000, 0.000]: no clear difference.

4. Findings and limitations

- Improvements are kept only where the interval supports them; otherwise the simpler model stays.
- Labels for overlap and flags are LLM-annotated reference labels (two passes, majority with a third). No human annotators were used; results are indicative. For 94 of 200 pairs the second labeller was a Claude prompt rather than Phi-3, so agreement on those items is not independent. Detector flags (Week 3 onwards) were labelled by two Claude passes with different wording plus a pass 3, with no Phi-3, so agreement on flags is not independent.
- Future work: bge-base prerequisite re-tuning not run (time).
- Data: NMIMS semester syllabus booklets; only 531 of 2,413 PDFs have a text layer (no OCR yet), so programme structures have gaps.
- Not repeated from the first Week 4: course search, LLM summaries and LLM prerequisite extraction, which do not serve the core objective directly.

AI assistance

Code and report text were written with AI assistance (Claude). Reference labels are LLM-annotated (Claude, Phi-3). Every number is generated from `results/*.json` by `tools/build_report.py` and verified by `tools/check_reports.py`.